



Anti-apoptotic effect of HeidihuangWan in renal tubular epithelial cells via PI3K/Akt/mTOR signaling pathway

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ABSTRACT

Ethnopharmacological relevance: Heidihuang Wan (HDHW) is a classic Chinese herbal formula, which was first recorded in the “Suwen Bingji Qiyi Baoming Collection” written by Liu Wansu during the Jin Dynasty (1115–1234 AD). It is commonly used clinically for the treatment of kidney diseases and its curative effect is stable. Previous animal experiments have confirmed that HDHW can effectively improve renal fibrosis. However, the underlying pharmacological mechanism remains unclear.

Aims of this study: Renal tubular epithelial cell (RTEC) apoptosis is one of the main pathological features of renal fibrosis. This study aimed to observe the effect and underlying mechanism of HDHW on the apoptosis of RTECs to further explore the pathological mechanism of HDHW against renal fibrosis.

Materials and methods: We examined the HDHW composition in rat serum. In vitro, we first screened out the optimal intervention concentration of HDHW on RTECs using the MTT assay. Hypoxia/reoxygenation was then used to induce apoptosis of RTECs (H/R-RTECs), which were divided into H/R-RTEC, astragaloside IV (positive control), HDHW, and RTECs groups. After 48 h of drug intervention, apoptosis of RTECs was detected using flow cytometry and protein expression was detected by western blotting. The 5/6 nephrectomy rat model was constructed and divided into the normal control, 5/6 nephrectomy, HDHW, and astragaloside IV groups. After 8 weeks of treatment, TUNEL staining was used to detect cell apoptosis, and western blotting was used to detect protein expression.

Results: HDHW downregulated the expression of pro-apoptotic proteins Bax and Caspase3, up-regulated the expression of anti-apoptotic protein Bcl-2, activated the PI3K/Akt/mTOR signaling pathway, and reversed the early apoptosis of RTECs, thereby resisting the apoptosis of RTECs.

Conclusion: HDHW inhibits apoptosis of RTECs by modulating the PI3K/Akt/mTOR signaling pathway. This study provides experimental evidence for the anti-fibrotic effect of HDHW on the kidneys and partially elucidates its pharmacological mechanism of action.

1. Introduction

Renal fibrosis is a common pathological process in various chronic kidney diseases (CKD) (Liu et al., 2017). Its occurrence and deterioration can accelerate kidney damage, hinder renal tissue repair, and eventually

lead to irreversible renal failure (da Silva et al., 2015), resulting in a heavy medical burden worldwide. HeidihuangWan (HDHW) has been widely used for the treatment of chronic renal failure caused by a variety of chronic kidney diseases, and it is safe and stable (Zhao and Zhang, 2017; Dong et al., 2013). We previously demonstrated that HDHW can

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Table 1
Specific information of HeidihuangWan decoction.

Chinses Name	Latin Name	Plant part used	Batch number	Dosage (grams)
Shu Di Huang (熟地黄)	<i>Rehmannia glutinosa</i> (Gaertn.) DC.	Root	19051981	8
Cang Zhu (苍术)	<i>Atractylodes carlinoides</i> (Hand.-Mazz.) Kitam.	Rhizome	19070691	6
Gan Jiang (干姜)	<i>Zingiber albiflorum</i> R. M.Sm.	Rhizome	19052411	0.5
Da Zao (大枣)	<i>Ziziphus jujuba</i> Mill.	Fruit	19060131	4

effectively improve renal fibrosis in 5/6 nephrectomized rats (Li, Y.Y. et al., 2022). However, the pharmacological mechanism of HDHW in renal fibrosis remains unclear, and this study aimed at exploring the underlying mechanism.

Apoptosis is an important factor that triggers the development and progression of renal fibrosis (Zhou and Liu, 2016; Chen et al., 2021). Apoptosis can trigger an inflammatory response, inflammatory cell infiltration, cytokine release, increased extracellular matrix (ECM) synthesis, and interstitial deposition, leading to renal fibrotic lesions (Chen et al., 2021), which in turn can cause further damage to RTECs. Inhibiting or delaying the apoptosis of RTECs can alleviate renal fibrosis (Jin et al., 2022; Chen, J. et al., 2022). Therefore, improving apoptosis of RTECs is a promising target for renal fibrosis intervention. Several traditional Chinese medicinal compounds (Zhao et al., 2022; Chen, Z. et al., 2022; Chen, R. et al., 2022) and their active ingredients (Madbouly et al., 2021; Shukla et al., 2018; Ghule et al., 2012) are effective in inhibiting RTECs apoptosis, thus becoming a promising treatment method for renal fibrosis. From the above, it can be seen that the improvement in renal tubular epithelial cell apoptosis is an important link in renal fibrosis. Several in vivo and in vitro studies have confirmed that the PI3K/Akt/mTOR signaling pathway plays an active role in regulating RTECs apoptosis and renal fibrosis (Liao et al., 2021; Li et al., 2020; Chen et al., 2020; An et al., 2020). Our previous studies confirmed that HDHW can resist renal fibrosis (Li, Y.Y. et al., 2022), and whether RTECs apoptosis is an important link in this process deserves further study.

Therefore, based on previous research, we propose the following hypothesis: HDHW can inhibit the apoptosis of RTECs by regulating the PI3K/Akt/mTOR signaling pathway, thereby improving renal fibrosis. This study is a further development of previous research.

2. Materials and methods

2.1. Drugs and reagents

All herbs used in this study were purchased from the Affiliated Hospital of Shandong University of Traditional Chinese Medicine (Jiangyin Tianjiang Pharmaceutical Co. Ltd. Jiangsu, China) and were quality monitored. Fourteen chemical constituents (linoleic acid, oleo-anolic acid, nicotinamide, rutin, gallic acid, shogaol, ferulic acid, scopoletin, 5-Hydroxymethyl-2-furaldehyde, coumarin, theophylline, 4-Coumaric acid, fraxetin, and trans-acnethole) were quantitatively analyzed according to our previous study using Ultra High Performance Liquid Chromatography-Quadrupole-Electrostatic Field Orbitrap Tandem Mass Spectrometry (UHPLC-Q-Orbitrap-MS/MS) (Li, Y.Y. et al., 2022). See Table 1 for specific information on herbal medicines. Plant names were checked on <http://www.theplantlist.org>, and the site was accessed on September 3, 2022.

Astragaloside IV with a purity of $\geq 98\%$ (HPLC) (batch number: SND-313) and carboxymethyl cellulose (Carboxymethyl Cellulose, CMC) (batch number: SND-5103) were purchased from Chuzhou Shinuoda Biotechnology Co., Ltd. Protein concentration assay kit (P0010S) was purchased from Shanghai Biyuntian Biotechnology Co., Ltd. Antibodies

against PI3K (4249), Akt (4691), and p-Akt (4060) were purchased from Cell Signaling Technology (Danvers, MA). Antibodies against p-mTOR (80596-1-RR) were purchased from Proteintech (Wuhan, China). Antibodies against mTOR (ab134903), Bax (ab32503), Bcl-2 (ab196495), Caspase3 (ab184787) were purchased from Abcam (Cambridge, United Kingdom). Horseradish enzyme-labeled goat anti-rabbit IgG (H + L) was purchased from Beijing Zhongshan Golden Bridge Biotechnology Co., Ltd. (Beijing, China). The FITC Annexin V Apoptosis Detection Kit with PI (556547) was purchased from BioLegend (San Diego, CA, United States). The TUNEL Detection Kit (G1507) was purchased from Wuhan Sevier Biotechnology Co., Ltd. (Wuhan, China). Dulbecco's modified Eagle's medium/nutrient mixture F-12 (DMEM/F-12) (11330032) was purchased from Thermo Fisher Scientific (China) Co., Ltd. Special culture medium for rat renal tubular epithelial cells (CM-R062) was purchased from Wuhan Procell Life Technology Co., Ltd.

2.2. Preparation of medicinal decoction

Assuming that an adult with a body weight of 60 kg takes 18.5 g HDHW per day, the dosage in rats is 1.94 g/kg according to the body surface area conversion factor of 6.3 between humans and rats. Our gavage dose was 3 mL/kg, so the HDHW concentration was 0.65 g/mL, which has been shown to be the optimal intervention dose of HDHW to improve renal fibrosis in 5/6 nephrectomized rats (Li, Y.Y. et al., 2022). The dosage of astragaloside IV (AS-IV) in rats was 1.5 mg/kg (Zhu et al., 2014). According to this ratio, AS-IV powder was dissolved in 0.3% CMC to prepare AS-IV decoction.

2.3. Preparation and component detection of medicated serum

All animals were purchased from the Beijing Weitong Lihua Laboratory Animal Technology Co., Ltd. (SCXK, 2016-0006). All animal experiments were conducted in compliance with the National Institute of Health Guide for the Care and Use of Laboratory Animals and approved by the Animal Ethics Committee of Shandong University of Traditional Chinese Medicine (SDUTCM20201019002).

Medicated serum was obtained from 7-week-old male Wistar rats as in previous studies (Gu et al., 2022; Huang et al., 2022; Yin et al., 2013). After 7 days of adaptive feeding, the mice were randomly divided into the HDHW, AS-IV, and blank groups. The HDHW and AS-IV groups were administered the corresponding decoction respectively, and the blank group was administered an equal volume of 0.9% normal saline at the same time every day. Rats were gavaged once daily for 10 days. Two hours after the last gavage, the rats were anesthetized by intraperitoneal injection of sodium pentobarbital (50 mg/kg) and blood was collected from the abdominal aorta. This was placed at room temperature for 2 h, centrifuged at 3000 rpm for 20 min at 4 °C, serum was separated, mixed with the same group of serum. The complement was inactivated in a water bath at 56 °C for 30 min, filtered and sterilized with a 0.22 µm filter, and then finally packed and frozen at -80 °C for later use.

Next, 1000 µL of 80% methanol was added to 200 µL of the solution of medicated serum of HDHW and vortexed for 10 min. This was then centrifuged at 12000 rpm for 10 min at 4 °C. The supernatant was filtered and analyzed on the machine, for details, see" (Li, Y.Y. et al., 2022)".

2.4. In vitro experiments

2.4.1. Extraction and identification of RTECs

In this study, we extracted primary RTECs from rats according to the literature (Wang et al., 2021) as follows. Male Wistar rats aged 2–4 weeks (Huagen et al., 2022) were selected, fasted for 12 h before the experiment, and had free access to water. Rats were euthanized by cervical dislocation and after soaking in 75% alcohol for 5 min, both kidneys were removed under sterile conditions and placed in PBS at 4 °C. The renal pedicle and capsule were removed, and a thin layer of the

renal cortex was cut and minced to a size of 1–3 mm³ (in PBS at 4 °C), and the tissue was transferred to a 15 mL centrifuge tube. PBS was added and the mixture was centrifuged three times at 500 rpm at room temperature for 3 min each time. After centrifugation, the supernatant was discarded, 2 mL of collagenase type II with a final concentration of 1 g/L was added to the obtained pellet, and the mixture was water bathed at 37 °C for 15 min. During the digestion process, it was gently shaken once every 3 min for 1 min. After digestion, centrifugation, and washing three times with PBS, 6 mL of DMEM/F12 medium was added, and the tissue was grinded through 80-mesh and 150-mesh cell sieves, filtered, and centrifuged at 800 rpm for 3 min at room temperature. The supernatant was discarded, the special culture medium was added, and the cells were mixed by pipetting several times to obtain a single-cell suspension. The cell suspension was inoculated into a cell culture flask and placed in a 37 °C, 5% CO₂ incubator for culture, and the flask was not shaken within 12 h. The medium was changed for the first time after 48 h, and was changed every other day thereafter. The 3rd to 5th passage cells were used for the experiments. For details, see the cell extraction flowchart (Fig. 2A).

Third-generation RTECs were used to prepare single-cell suspensions, and the cell density was adjusted to 1×10^5 cells/mL. The cell slides were placed in a 6-well plate, 300 µL of cell suspension was dropped into the center of the cell slides, placed in an incubator for 1 h, and then 2 mL of culture solution was added to the 6-well plate after the cells adhered to the wall. This was incubated for 48 h. When the cells were confluent to 70–80%, they were washed three times with PBS for 5 min each time on a shaker. The residual PBS was discarded, 1 mL of 4% paraformaldehyde added and fixed for 20 min at room temperature, and the slides were washed with PBS. The residual PBS was discarded, 200 µL 5% BSA blocking solution added and blocked at room temperature for 20 min. The blocking solution was then discarded. A total of 200 µL cytokeratin (CK18) antibody (1:1000, rabbit anti-rat), was added to replace the negative control group with PBS and incubated overnight at 4 °C. The primary antibody was removed the next day, the cells were washed three times with PBS for 5 min each time, and the residual PBS was discarded. A total of 200 µL of FITC-labeled goat anti-rabbit IgG was added, incubated at room temperature for 1 h in the dark, the secondary antibody discarded and washed 3 times with PBS, 5 min each time, and the residual PBS discarded. A total of 200 µL of DAPI staining solution was added, stained for 3 min at room temperature in the dark, washed with PBS three times, 5 min each time, and the residual PBS discarded. A drop of PBS was added on the coverslip, the cell slide removed from the well plate, the slide covered with the cell side down, and the cover slip tapped with the blunt end of the tweezers to remove air bubbles. Fluorescence labeling was observed using a fluorescence microscope.

2.4.2. MTT assay to detect the optimal intervention concentration of HDHW-containing serum on RTECs

The RTECs of the P3 generation that grew and fused to 80–90% were collected and divided into the control, HDHW drug-containing serum group (1%, 5%, 10%, 15%, 20%), and AS-IV drug-containing serum groups (1%, 5%, 10%, 15%, 20%). There were three replicate wells per group. HDHW drug-containing serum group and AS-IV drug-containing serum group were added to 100 µL of culture medium containing the corresponding concentration of drug-containing serum to each well, and the control group was added with the same concentration of blank drug-containing serum culture medium. After 48 h, 20% MTT solution was added, incubated at 37 °C for 3 h in the dark, 150 µL of DMSO solution added after discarding the solution, and the absorbance measured at 495 nm (A) using a microplate reader. Proliferation rate calculation formula: [(average absorbance of drug group-average absorbance of the control group)/average absorbance of control group $\times$ 100%].

2.4.3. Construction of apoptotic cell model and drug intervention

The adherent RTECs were placed in an Anaero culture bag (92%N₂ + 3%O₂ + 5%CO₂), cultured in a sugar-free DMEM medium for 12h, and

then transferred to a 5% CO₂ incubator (95%O₂ + 5%CO₂), replaced with a low-glucose DMEM medium containing 10% FBS, and continued to culture for 12 h to obtain apoptotic RTECs (H/R-RTECs) (Yuan et al., 2017; Jiang et al., 2017).

H/R-RTECs were divided into H/R-RTECs, AS-IV, HDHW, and RTECs groups without apoptosis induction. The HDHW and AS-IV groups were, respectively, added with the corresponding optimal concentration of drug-containing serum culture medium, and the RTECs and H/R-RTEC groups were, respectively, added with the same concentration of blank drug-containing serum culture medium. After 48 h of culture, apoptosis was analyzed.

2.4.4. Detection of apoptosis by flow cytometry

Cells were washed twice with cold PBS and resuspended in Annexin V Binding Buffer at a concentration of 1×10^6 cells/mL. A total of 100 µL of the cell suspension was transferred to a 5 mL test tube. Also, 5 µL of FITC Annexin V. and 10 µL of propidium iodide solution were added. The cells were gently vortexed and incubated for 15 min at room temperature (25 °C) in the dark. Next, 400 µL of annexin V binding buffer was added to each tube. Analysis was made by flow cytometry with appropriate machine settings.

2.4.5. Western blot analysis

After intervention, the cells were washed once with PBS and 150 µL of the cell lysate was added to each well. Cells were collected using a cell scraper and placed in ice water for further lysis for 20 min. After lysis, it was centrifuged at 12,000 rpm for 15 min at 4 °C and the supernatant collected for protein quantification. After protein concentration detection, sample addition, gel electrophoresis, membrane transfer, blocking with 5% nonfat milk powder, primary antibody incubation (anti-Bax 1:2000, anti-Bcl-2 1:1000, anti-Caspase3 1:2000, anti-P13K 1:1000, anti-Akt 1:1000, anti-mTOR 1:10000, anti-p-Akt 1:2000, anti-p-mTOR 1:10000), membrane washing, secondary antibody incubation, membrane washing, and finally adding developer solution to develop were done. The specific steps were the same as those in our previous study “(Li, Y.Y. et al., 2022)”.

2.5. In vivo experiments

2.5.1. Animal model establishment and drug intervention

In this study, 7-week-old male Wistar rats (200 $\pm$ 20 g) were selected as research subjects. After successfully constructing the 5/6 nephrectomy rat model according to “(Li, Y.Y. et al., 2022)”, we randomly divided the 5/6 nephrectomy rats into the 5/6 nephrectomy group, AS-IV group, HDHW group, and another normal control (NC) group, a total of four groups (n = 6). The HDHW and AS-IV groups were administered the corresponding liquids. The normal control group and 5/6 nephrectomy group were administered the same volume of normal saline. After continuous administration for 8 weeks, the rats were anesthetized by intraperitoneal injection of sodium pentobarbital (50 mg/kg) and the kidneys were immediately separated. The specific process is illustrated in Fig. 5A.

2.5.2. TUNEL staining

This was dewaxed to water. The proteinase K working solution was added to cover the tissue and incubated at 37 °C for 20 min. The cells were shaken and washed three times in PBS for 5 min each time. The tissue was covered with a membrane-breaking working solution, incubated at room temperature for 20 min, and washed thrice in PBS with shaking for 5 min each time. The sections were placed in a 3% hydrogen peroxide solution, incubated at room temperature for 20 min in the dark, and washed three times in PBS with shaking for 5 min each time. Buffer was added to cover the tissue and incubated at room temperature for 10 min. The equilibration solution was removed, mixed according to the ratio of Recombinant TdT enzyme: Biotin-dUTP Labeling Mix: Equilibration Buffer = 1 µL: 5 µL: 50 µL, and added dropwise to the slice

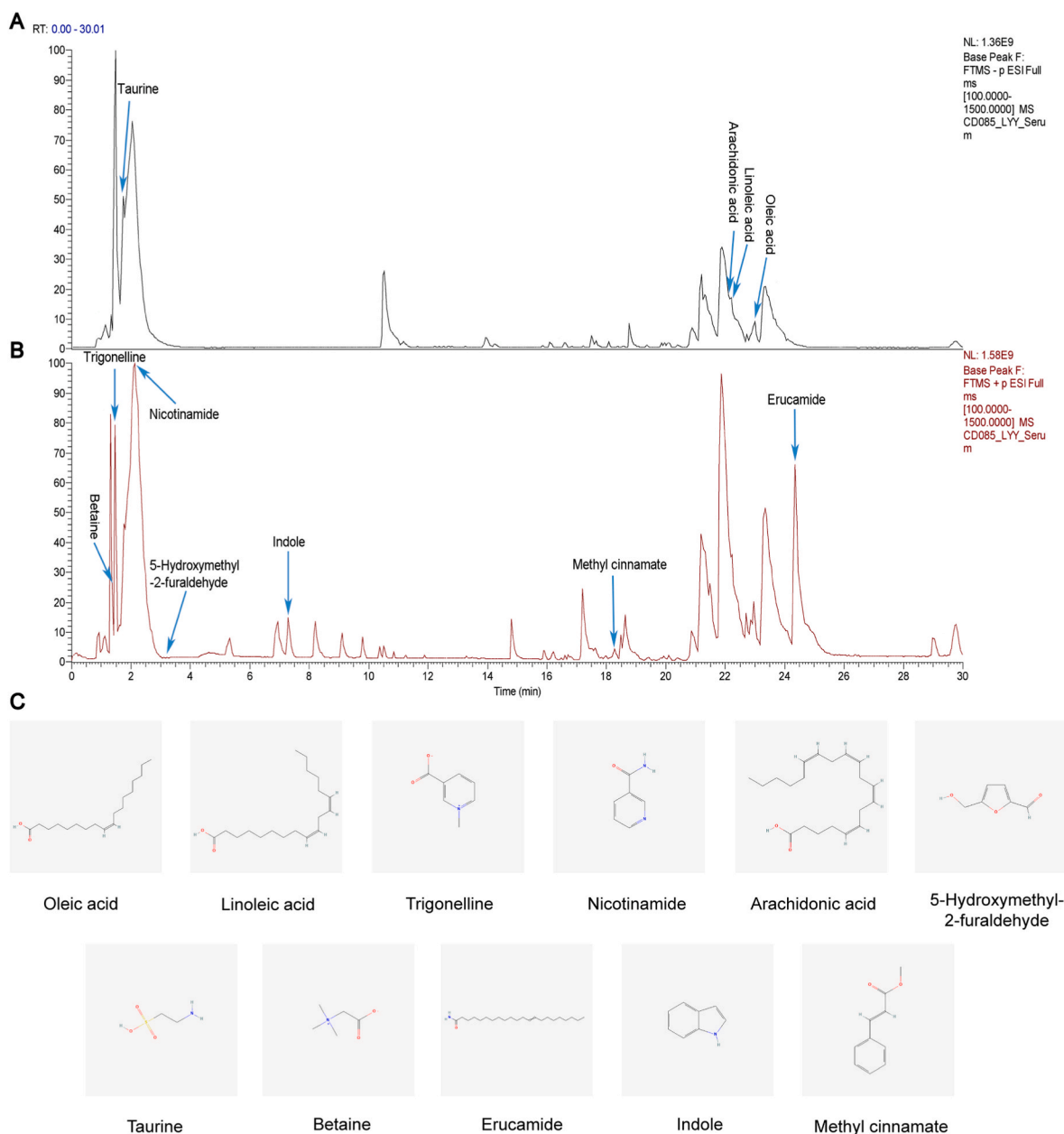


Fig. 1. Identification of chemical components from the medicated serum of HeidihuangWan (HDHW). Total ion chromatography in the negative (A) and positive (B) ion modes for medicated serum of HDHW samples are shown. (C) Molecular structure of constituents.

to cover the tissue. The slice was then placed flat in a humid box, and incubated at 37 °C for 1 h, with a small amount of water added to the wet box to keep it moist. The cells were shaken and washed three times in PBS for 5 min each time. Streptavidin-HRP and TBST were mixed at a ratio of 1:200, dripped to cover the tissue, sliced flat in a humid box, and incubated in at 37 °C incubator for 30 min. The cells were shaken and washed in PBS three times for 5 min each. Freshly prepared DAB chromogenic solution was added, and the chromogenic time was controlled under the microscope. The positive signal was that the nucleus was brownish yellow, and then rinsed with pure water to stop chromogenic development. Hematoxylin was counterstained for 45 s, washed with pure water, differentiated with differentiation solution for several seconds, washed with pure water, returned to blue with ammonia, and rinsed with pure water. Conventional dehydration and transparent mounting medium Hematoxylin-stained nuclei are blue, and DAB-positive apoptotic nuclei are brown.

2.5.3. Western blot analysis

The frozen kidney tissue was weighed, and 500 µL tissue lysate (containing 5 µL PMSF, 5 µL phosphatase inhibitor) was added to each 50 mg kidney tissue and homogenized in a tissue homogenizer. The homogenized samples were further lysed in ice water for 20 min and centrifuged at 12,000 rpm for 15 min at 4 °C. The supernatant was collected and stored at 4 °C until later use. The remaining steps are the same as those described in Section “2.4.5”. The primary antibodies used for incubation were: anti-Bax 1:2000, anti-Bcl-2 1:1000, anti-Caspase3 1:2000.

2.6. Statistical analysis

To ensure the accuracy of the experimental results, all experiments in this study were independently repeated three times, and all statistical analyses were performed using SPSS 26.0 statistical software (IBM, United States). Statistical analysis was performed using one-way

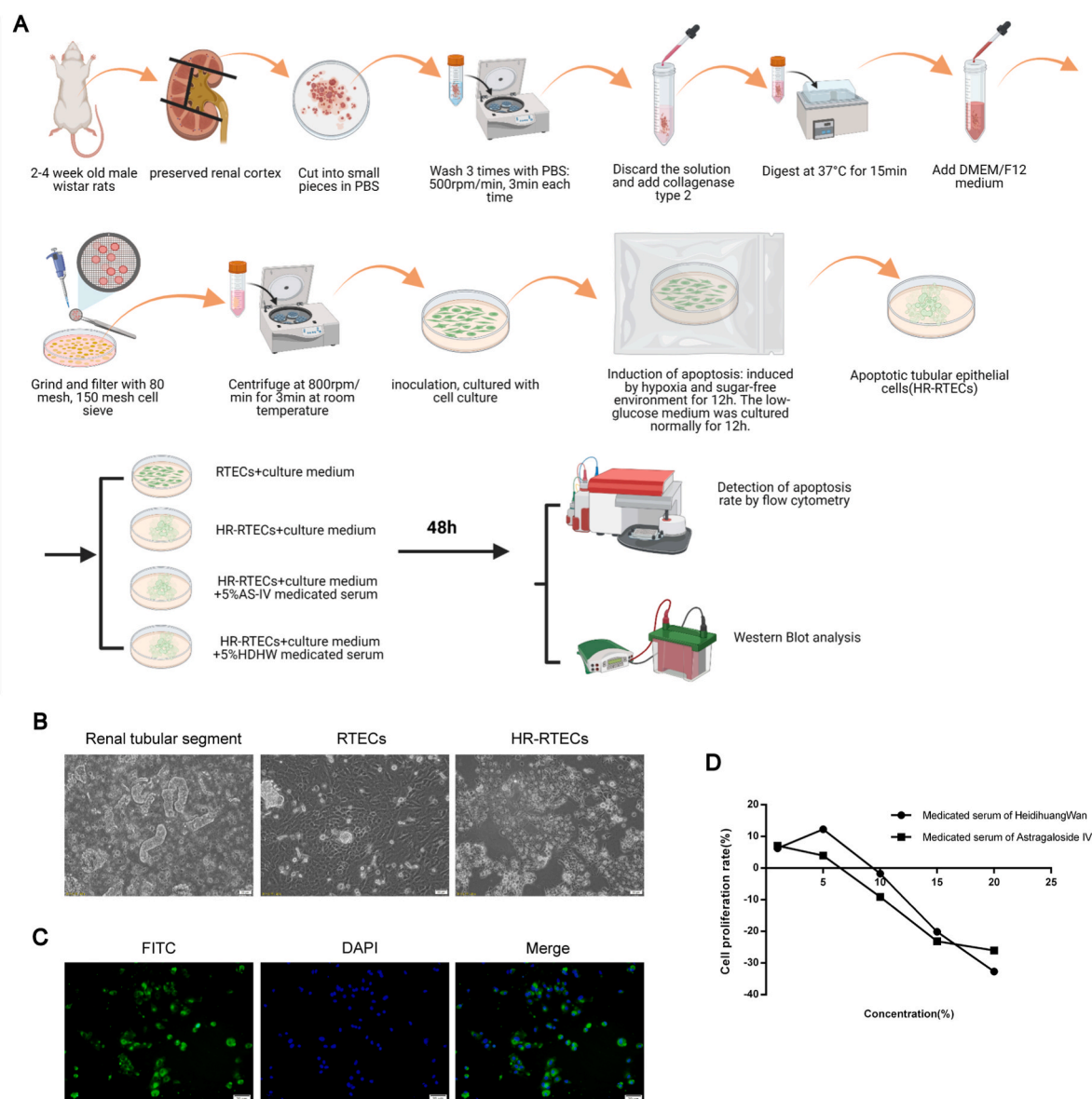


Fig. 2. 5% HDHW-containing serum is the optimal concentration for intervention of RTECs derived from rats. (A) Flow chart of the cell experiment. (B) Cell morphology of the extracted RTECs under the ordinary light microscope ($\times 200$). (C) Immunofluorescence was used to identify the expression of FITC-CK18 in RTECs (Advanced Upright Fluorescence Microscopy, $\times 200$). Green fluorescence represented FITC-labeled CK18, and blue fluorescence represented nuclei. (D) MTT was used to examine the effect of different concentrations of drug-containing serum on the proliferation of RTECs (Detection on microplate reader). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

analysis of variance (ANOVA), followed by the LSD test or Dunnett T3 test. All data are presented as the mean $\pm$ standard deviation. Results were considered statistically significant at $p < 0.05$. Graphs were constructed using GraphPad Prism 6.0 (GraphPad Software, United States).

3. Results

3.1. Component analysis of the drug-containing serum of HeidihuangWan

The major compounds in the medicated serum sample of HDHW were identified using UHPLC-Q-Orbitrap-MS/MS analysis. The total negative (Fig. 1A) and positive (Fig. 1B) ion chromatograms of medicated serum of HDHW demonstrated the chemical composition of all the compounds. Several compounds were detected in the medicated serum of HDHW. As shown in Figs. 1C and 11 compounds were identified: oleic acid, linoleic acid, trigonelline, nicotinamide, arachidonic acid, 5-Hydroxymethyl-2-furaldehyde, taurine, betaine, erucamide, indole,

and methyl cinnamate.

3.2. 5% HeidihuangWan-containing serum is the optimal concentration for intervention of RTECs derived from rats

We first extracted RTECs from rat kidneys in vitro and observed their cell morphology. The results showed that the cells extracted from the rat renal cortex grew adherently, with a cobblestone-like distribution and compact arrangement, which was consistent with the morphological appearance of renal tubular epithelial cells (Fig. 2B). CK18 is a specific marker of RTECs (Xu et al., 2021), and our extracted cells expressed CK18 after immunofluorescence labeling (Fig. 2C). These results indicated that the extracted cells were RTECs. To determine the optimal dose of drug-containing serum for intervention in RETCs, we performed an MTT assay and calculated the proliferation rate of RTECs. The results showed that the HDHW drug-containing serum concentration of 5% was most beneficial to the proliferation of RTECs, and 5% AS-IV

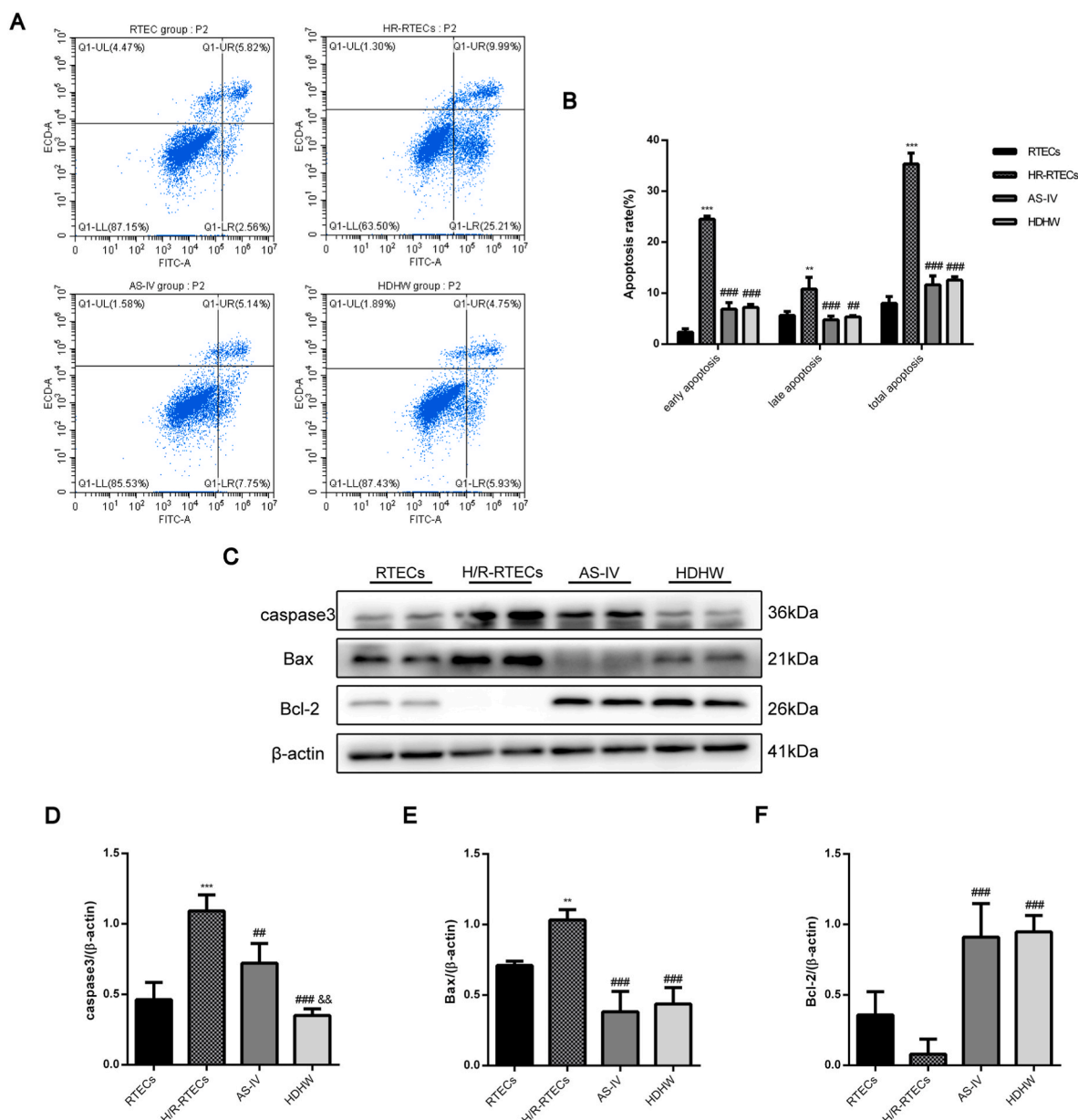


Fig. 3. The drug-containing serum of HeidihuangWan (HDHW) inhibits the apoptosis of RTECs in vitro. (A) FITC Annexin V Apoptosis Detection Kit with PI was used to label apoptotic cells after drug intervention, and flow cytometry detects labeled apoptotic cells. The upper right quadrant is late apoptotic cells, the lower right quadrant is early apoptotic cells, the upper left quadrant is necrotic cells, and the lower left quadrant is living cells. (B) Statistical analysis of apoptosis rate. (C) Western Blotting to detect the protein levels of Caspase3, Bax, Bcl-2. (D, E, F) Protein concentration analysis. The data are expressed as ** $p < 0.01$, *** $p < 0.001$: compared with RTECs group; ## $p < 0.01$, ### $p < 0.001$: compared with H/R-RTECs group; && $p < 0.01$: compared with AS-IV (positive control) group.

drug-containing serum could also promote the proliferation of RTECs, so we chose 5% as the intervention concentration of drug-containing serum (Fig. 2D).

3.3. The drug-containing serum of HeidihuangWan inhibits the apoptosis of RTECs in vitro

Apoptosis of RTECs has been recognized as a key to the pathogenesis of renal fibrosis (He et al., 2022). AS-IV was found to be significantly effective in inhibiting the apoptosis of RTECs (Xu et al., 2014; Wang et al., 2020; Ju et al., 2019; Chen et al., 2018). Therefore, we compared the antiapoptotic effects of HDHW and AS-IV in RTECs. As shown in Fig. 3, the results of flow cytometry showed that the early apoptosis rate, late apoptosis rate, and total apoptosis rate of the H/R-RTEC group were significantly higher than those of the RTECs group ($p < 0.01$); HDHW

and AS-IV reversed the occurrence of early apoptosis in RTECs ($p < 0.001$), reduced the number of cells in the late stage of apoptosis ($p < 0.01$), and reduced the total apoptotic rate of RTECs ($p < 0.001$). At the same time, there was no significant difference in the apoptosis rate between the HDHW group and the AS-IV group ($p > 0.05$). In addition, western blot analysis showed that the expressions of Caspase3 and Bax in the H/R-RTEC group was significantly higher than that in the RTECs group ($p < 0.01$), but the expression of Bcl-2 was not statistically significant between the two groups. HDHW and AS-IV significantly downregulated the expression of Caspase3 and Bax ($p < 0.01$), and significantly upregulated the expression of Bcl-2 ($p < 0.001$). Moreover, the inhibition of Caspase3 expression by HDHW was greater than that of the positive control drug AS-IV ($p < 0.01$).

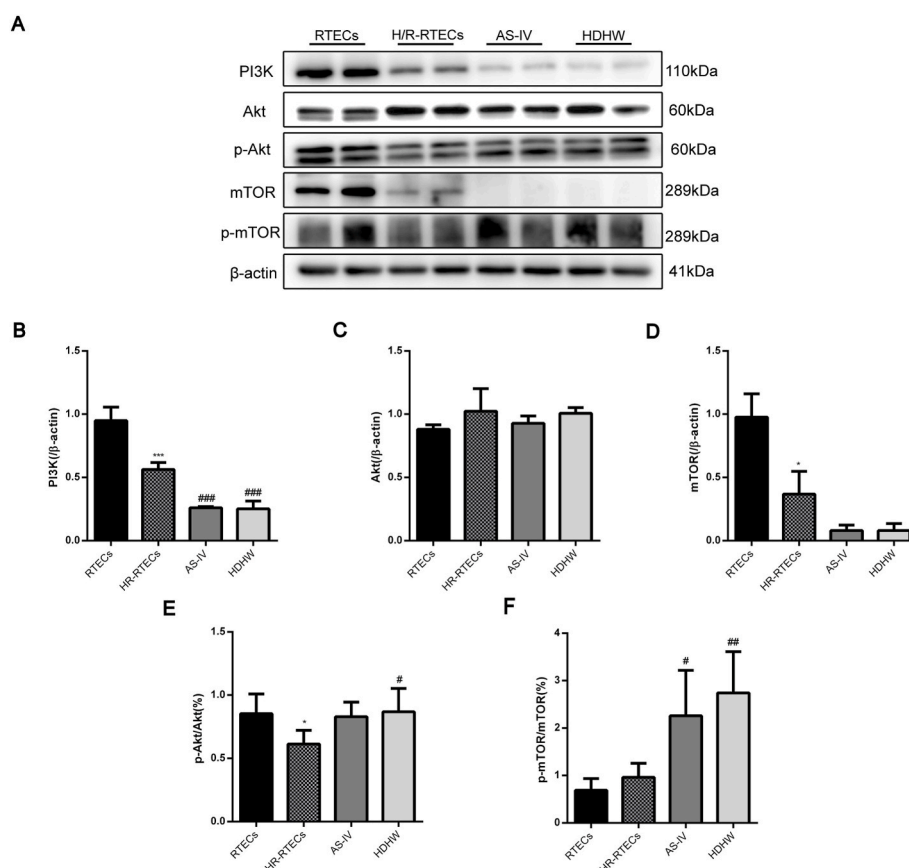


Fig. 4. HeidihuangWan (HDHW) activates PI3K/Akt/mTOR signaling pathway in RTECs. (A) Western Blotting to detect the protein levels of PI3K, Akt, p-Akt, mTOR, and p-mTOR. (B, C, D, E, F) Protein concentration analysis. The data are expressed as * $p < 0.05$, *** $p < 0.001$: compared with the RTECs group; # $p < 0.05$, ## $p < 0.01$, ### $p < 0.001$: compared with the H/R-RTECs group.

3.4. HeidihuangWan activates PI3K/Akt/mTOR signaling pathway in RTECs

PI3K/Akt/mTOR is a key signaling pathway that regulates cell proliferation, apoptosis, and differentiation (Aggarwal et al., 2022). Activation of the PI3K/Akt/mTOR signaling pathway can inhibit renal cell apoptosis and improve renal fibrosis (Zhang et al., 2019; Yang et al., 2022). As shown in Fig. 4, PI3K, mTOR, and p-Akt/Akt were significantly downregulated in the H/R-RTEC group compared to the RTECs group ($p < 0.05$). There were no significant differences in the expression of Akt and p-mTOR/mTOR between the two groups ($p > 0.05$). However, the drug-containing serum of HDHW significantly upregulated the expression of p-Akt/Akt and p-mTOR/mTOR compared to the H/R-RTEC group ($p < 0.05$).

3.5. HeidihuangWan inhibits apoptosis of cells in kidney tissue

As shown in Fig. 5, the results of TUNEL staining showed that the number of apoptotic cells in the 5/6 nephrectomy group was significantly increased, compared with the normal control group ($p < 0.001$); Both HDHW and AS-IV significantly reduced the number of apoptotic cells in renal tubules, compared with the 5/6 nephrectomy group ($p < 0.001$). Simultaneously, we used western blotting to analyze the expressions of Caspase3, Bax, and Bcl-2. The results showed that in the 5/6 nephrectomy group, the expression of Caspase3 was higher than that in the normal control group ($p < 0.05$), while the expression of Bcl-2 was significantly lower than that in the normal control group ($p < 0.01$), and there was no statistically significant difference in the expression of Bax between the two groups ($p > 0.05$). HDHW and AS-IV significantly downregulated the expression of Caspase3 and Bax ($p < 0.05$) and

upregulated the expression of Bcl-2 ($p < 0.05$), compared with the 5/6 nephrectomy group.

4. Discussion

In this study, we analyzed the main active components of HDHW after analyzing the blood of rats to identify the possible targets and mechanisms of HDHW's nephroprotective effects. UHPLC-Q-Orbitrap-MS/MS analysis showed that the drug-containing serum of HDHW contained 11 chemical components. Oleic acid (Zhang, B. et al., 2022; Lee et al., 2021), linoleic acid (Ogborn et al., 2003; Zhan et al., 2015), Trigonelline (Ghule et al., 2012; Peerapen and Thongboonkerd, 2020; Chen, C. et al., 2022), Nicotinamide (Kumakura et al., 2021), arachidonic acid (Turolo et al., 2021), 5-Hydroxymethyl-2-furaldehyde (Shukla et al., 2018), taurine (Madbouly et al., 2021; Huang et al., 2008; Karbalay-Doust et al., 2012), betaine (Sharma et al., 2021), indole (Qin et al., 2022), and methyl cinnamate (Zhu et al., 2017) have been confirmed to have nephroprotective effects. Among these, arachidonic acid (Turolo et al., 2021), linoleic acid (Ogborn et al., 2003), trigonelline (Ghule et al., 2012; Peerapen and Thongboonkerd, 2020), taurine (Madbouly et al., 2021; Huang et al., 2008; Karbalay-Doust et al., 2012), and Indole (Qin et al., 2022), these five components have significant anti-renal fibrosis effects. Indole in particular, can alleviate fibrosis in the lung, myocardium, kidney, and liver by inhibiting the inflammatory response, oxidative stress, TGF- β /Smad pathway, and other signaling pathways, which are important candidates for organ fibrosis therapy (Qin et al., 2022). At the same time, Trigonelline (Ghule et al., 2012), Taurine (Madbouly et al., 2021), and 5-Hydroxymethyl-2-furaldehyde (Shukla et al., 2018) can inhibit renal cell apoptosis, downregulate the expression of pro-apoptotic proteins Caspase3 and Bax, and upregulate

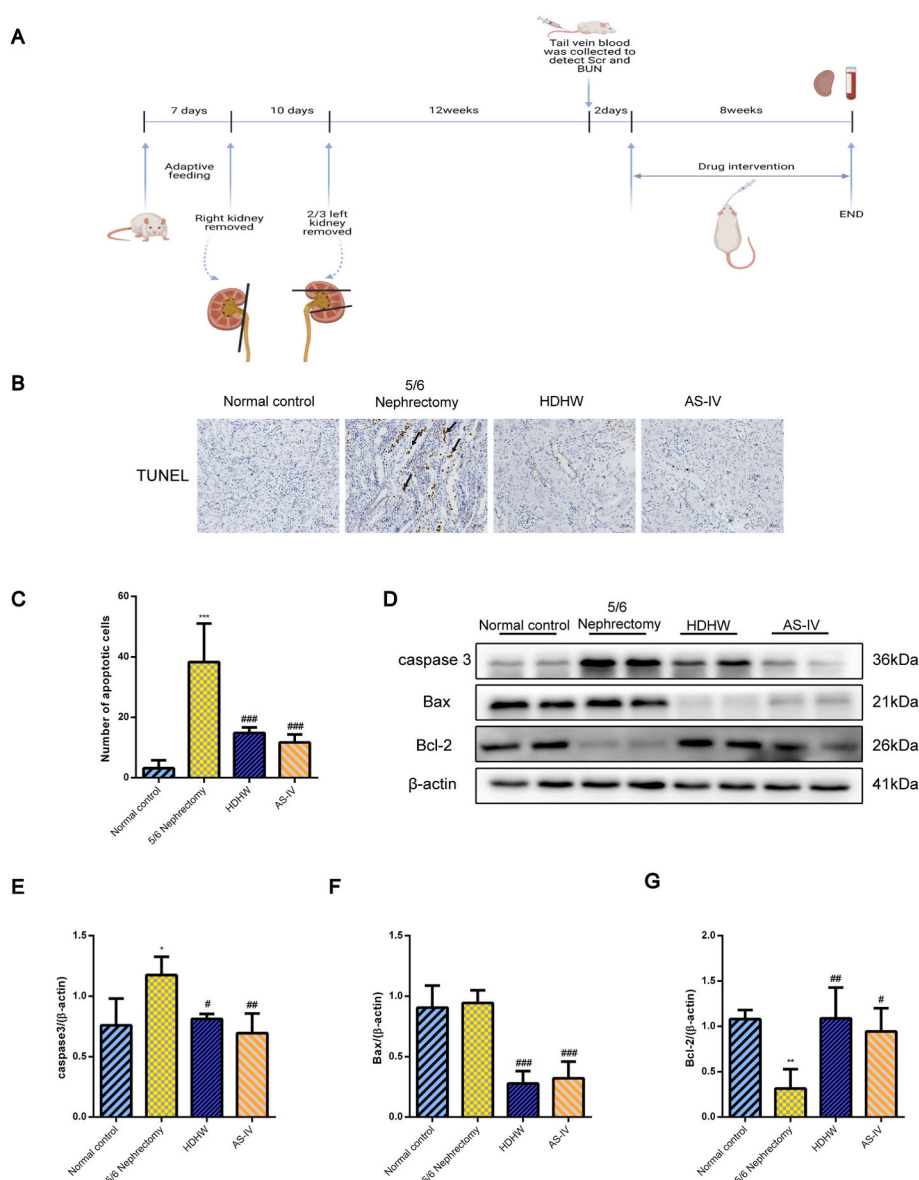


Fig. 5. HeidihuangWan (HDHW) inhibits cell apoptosis in kidney tissue. (A) Animal Experiment Timeline. (B) Apoptosis was detected by TUNEL staining. Positive signals for apoptosis were brownish yellow and were observed in the 5/6 nephrectomy group (arrows). (C) Image J software was used to statistically analyze the number of apoptotic cells. (D) Western Blotting to detect the protein level of Caspase3, Bax, and Bcl-2. (E, F, G) Protein concentration analysis. The data are expressed as * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$: compared with normal control group; # $p < 0.05$, ## $p < 0.01$, ### $p < 0.001$: compared with 5/6 nephrectomy group, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

the expression of anti-apoptotic protein Bcl-2 to improve renal fibrosis. Therefore, anti-renal fibrosis, regulation of renal cell apoptosis, inhibition of the inflammatory response, and oxidative stress may be the main targets of HDHW to protect the kidney. It should be noted that although many active ingredients in HDHW have been shown to have anti-fibrotic effects, the anti-fibrotic effects and targets of these active ingredients in the 5/6 nephrectomy rat model remain unclear. In addition, since rats were the research objects of this study, the drug-containing serum of HDHW of rats was selected for the analysis of drug components, but the active components of HDHW in human blood were not analyzed.

The main pathological process of renal fibrosis can be summarized as renal epithelial cell damage leading to fibroblast proliferation (Gewin, 2018) and macrophage infiltration (Peng et al., 2015). Animal models and pathological specimens of various kidney diseases have shown that regions with heavier interstitial fibrosis undergo more apoptosis in RTECs, and the degree of interstitial fibrosis is positively correlated with the number of RTECs (Zhou and Liu, 2016). In other words, renal tubular epithelial cells are the main target of various kidney injuries. Therefore, an in-depth study of the apoptotic mechanism of RTECs is an effective means to elucidate the pathological mechanism of renal fibrosis (Chen et al., 2021). In vitro, our results suggest that HDHW can

reverse the early apoptosis of RTECs induced by hypoxia/reoxygenation, thereby reducing the number of late apoptotic cells and improving the apoptosis of RTECs. Simultaneously, we detected proteins representing apoptosis to further confirm the inhibitory effect of HDHW on the apoptosis of RTECs. Bcl-2 inhibits apoptosis, and Bax promotes apoptosis. They are key factors involved in apoptosis (Zhi et al.). Caspase3 is the only apoptotic protease cascade that is an important effector of apoptosis (Yan et al., 2022). Our results suggest that Bcl-2, Bax, and Caspase3 are all targets of HDHW to regulate apoptosis in RTECs. HDHW inhibits the apoptosis of RTECs by up-regulating the expression of anti-apoptotic protein Bcl-2 and down-regulating the expression of pro-apoptotic proteins Caspase3 and Bax. In addition, we verified the anti-apoptotic effect of HDHW on renal tubular cells in vivo. TUNEL staining is an in situ terminal transferase labeling technique that can perform in situ staining of intact single apoptotic cell nuclei or apoptotic bodies and is a classic apoptotic cell detection technique. Therefore, we used TUNEL staining to label the nuclei of the apoptotic RTECs. In contrast, we detected the expression of apoptosis-related proteins in the kidney tissue of 5/6 nephrectomy rats using western blotting. Statistical analysis showed that HDHW reduced the number of apoptotic cells in renal tubules, upregulated the expression of anti-apoptotic protein

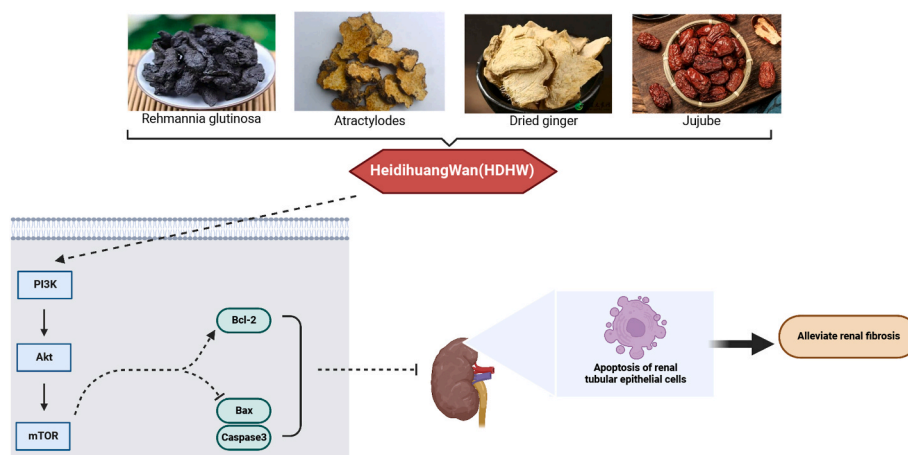


Fig. 6. Schematic diagram of the mechanism of HeidihuangWan inhibiting the apoptosis of RTECs.

Bcl-2, and downregulated the expression of pro-apoptotic proteins Caspase3 and Bax. Therefore, both in vivo and in vitro results confirmed that HDHW can inhibit the apoptosis of RTECs, and Bcl-2, Bax, and Caspase3 are important targets of HDHW that regulate the apoptosis of RTECs.

The PI3K/Akt pathway is an intracellular signaling pathway that plays an important regulatory role in cell growth, death, proliferation, and aging. It activates PI3K phosphorylation, further stimulates Akt localized in the cell membrane, and activates Akt to trigger various biological functions. mTOR is activated by PI3K and Akt (Liang et al., 2022) and regulates renal apoptosis (An et al., 2020; Wei et al., 2019). Our findings suggest that HDHW upregulates the expression of phosphorylated proteins in the PI3K/Akt/mTOR signaling pathway. The activation of this pathway may be the underlying mechanism by which HDHW inhibits the apoptosis of RTECs. However, the role of PI3K/Akt/mTOR in renal fibrosis remains controversial. Several studies have shown that activation of the PI3K/Akt/mTOR signaling pathway reduces apoptosis of RTECs and improves renal fibrosis (Liao et al., 2021; Xu et al., 2022; Zhang, J. et al., 2022; Zhang et al., 2021), which is consistent with our current findings. Other findings suggest that the improvement in RTECs apoptosis and renal fibrosis is related to the inhibition of the PI3K/Akt/mTOR signaling pathway (Xiao et al., 2021; Li, D. et al., 2022; Hu et al., 2021; He et al., 2020). Unfortunately, the lack of rescue validation relying on PI3K/Akt/mTOR pathway inhibitors in our study resulted in an incompletely reliable relationship between HDHW resistance to apoptosis in RTECs and the activation of the PI3K/Akt/mTOR pathway. Nevertheless, our study confirmed that HDHW can inhibit the apoptosis of RTECs, and that the PI3K/Akt/mTOR signaling pathway is activated during this process, which may be the pharmacological pathway of HDHW against renal fibrosis (Fig. 6).

5. Conclusion

In conclusion, our study shows that HeidihuangWan can effectively inhibit the apoptosis of renal tubular epithelial cells, accompanied by the activation of the PI3K/Akt/mTOR signaling pathway, which may be a potential mechanism by which HeidihuangWan improves renal fibrosis.

Ethics statement

The animal study was reviewed and approved by the Ethics Committee of Shandong University of Traditional Chinese Medicine.

Conflicts of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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CRediT authorship contribution statement

Ying-Ying Li: Conceptualization, Methodology, Investigation, Writing - original draft. **Zeng-Hui Tian:** Investigation, Formal analysis, Writing - original draft. **Shan-Shan Su:** Data Curation, Visualization. **Jing-Jing Shi:** Data Curation, Visualization. **Chao Zhou:** Investigation. **Li-Hua Zhang:** Investigation. **Fa-Rong Zhang:** Supervision, Funding acquisition. **Yan-Ke Hao:** Writing - review & editing, All authors have reviewed and approved the final version of the manuscript.

Declaration of competing interest

The authors declare no conflicts of interest.

Data availability

Data will be made available on request.

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List of abbreviations

full name abbreviation

HeidihuangWan HDHW

Renal tubular epithelial cells RTECs

Apoptotic renal tubular epithelial cells induced by hypoxia/

reoxygenation H/R-RTECs
 Astragaloside IV AS-IV
 Phosphatidylinositol-3 kinase/protein kinase B/mammalian target of rapamycin PI3K/Akt/mTOR
 Extracellular matrix ECM
 Chronic kidney disease CKD
 Terminal deoxynucleotidyl transferase-mediated dUTP-biotin nick end labeling assay TUNEL
 hypoxia/reoxygenation H/R

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